# POINTS OF A SITE AND POINTS OF ITS TOPOS

THE STACKS PROJECT AUTHORS

*Editorial note.* All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

*Editorial note.* The proof is the author's complete argument; the parenthetical 'verification omitted' at source line 8318 concerns the identification of two bijections on a representable sheaf, which the surrounding argument then uses and extends, and it is preserved verbatim inside the counted span.

*Editorial note.* Every labelled reference occurring inside the counted spans is carried in assembly.external_references as a non-load-bearing navigation pointer (same-chapter ones at their definition site, cross-chapter ones at the invocation inside the counted span), so no load-bearing mathematics is replaced by a stub. No bibliography citation occurs inside the counted spans.

*Editorial note.* The checked chapter list for same-name collisions is the set of stacks chapters staged in this repository: chow, cohomology, divisors, etale-cohomology, flat, more-algebra, more-morphisms, morphisms, spaces-more-morphisms, varieties together with this card's chapter sites-cohomology.

*Editorial note.* The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

*Allowed standard prerequisites (editorial).*

- The definition of a point of a site and of a topos. Same chapter, Definition definition-point and Definition definition-topos. Supply the conditions verified in both directions.
- Coverings become surjections after sheafification. Same chapter, Lemma lemma-covering-surjective-after-sheafification. Supplies the surjectivity used to prove the first condition of a point of a site.
- Sheafification commutes with finite limits. Same chapter, Lemma lemma-sheafification-exact. Supplies the commutation with fibre products used for the second condition.
- Every sheaf is a coequalizer of coproducts of representables. Same chapter, Lemma lemma-sheaf-coequalizer-representable. Allows the comparison map to be checked on representable sheaves and then extended through colimits.

*Source reference.* Original definition 00Y5: sources/stacks-sites/pinned-original-source.tex, source lines 8278–8278. Navigation pointer invoked at proof line 8278; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

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*Date*: stacks-project:sites.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

*Source reference.* Original definition 00XA: sources/stacks-sites/pinned-original-source.tex, source lines 8280–8280. Navigation pointer invoked at proof line 8280; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

*Source reference.* Original lemma 00WT: sources/stacks-sites/pinned-original-source.tex, source lines 8286–8286. Navigation pointer invoked at proof line 8286; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

*Source reference.* Original lemma 00WJ: sources/stacks-sites/pinned-original-source.tex, source lines 8292–8292. Navigation pointer invoked at proof line 8292; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

*Source reference.* Original equation 04EH: sources/stacks-sites/pinned-original-source.tex, source lines 8301–8301. Navigation pointer invoked at proof line 8301; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

*Source reference.* Original lemma 00Y6: sources/stacks-sites/pinned-original-source.tex, source lines 8308–8308. Navigation pointer invoked at proof line 8308; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

*Source reference.* Original lemma 00Y8: sources/stacks-sites/pinned-original-source.tex, source lines 8235–8235. Navigation pointer invoked at proof line 8235; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

*Source reference.* Original lemma 00WS: sources/stacks-sites/pinned-original-source.tex, source lines 8321–8321. Navigation pointer invoked at proof line 8321; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

**Lemma 0.1.** *Let $\mathcal{C}$ be a site.*

(1) *Let $p$ be a point of the site $\mathcal{C}$. Then the pair of functors $(p_*, p^{-1})$ introduced above define a morphism of topoi $Sh(pt) \to Sh(\mathcal{C})$.*
(2) *Let $p = (p_*, p^{-1})$ be a point of the topos $Sh(\mathcal{C})$. Then the functor $u : U \mapsto p^{-1}(h_U^\#)$ gives rise to a point $p'$ of the site $\mathcal{C}$ whose associated morphism of topoi $(p'_*, (p')^{-1})$ is equal to $p$.*

*Proof.* Proof of (1). By the above the functors $p_*$ and $p^{-1}$ are adjoint. The functor $p^{-1}$ is required to be exact by Definition 00Y5. Hence the conditions imposed in Definition 00XA are all satisfied and we see that (1) holds.

Proof of (2). Let $\{U_i \to U\}$ be a covering of $\mathcal{C}$. Then $\coprod (h_{U_i})^\# \to h_U^\#$ is surjective, see Lemma 00WT. Since $p^{-1}$ is exact (by definition of a morphism of topoi) we

conclude that $\coprod u(U_i) \to u(U)$ is surjective. This proves part (1) of Definition 00Y5. Sheafification is exact, see Lemma 00WJ. Hence if $U \times_V W$ exists in $\mathcal{C}$, then

$$h^\#_{U \times_V W} = h^\#_U \times_{h^\#_V} h^\#_W$$

and we see that $u(U \times_V W) = u(U) \times_{u(V)} u(W)$ since $p^{-1}$ is exact. This proves part (2) of Definition 00Y5. Let $p' = u$, and let $\mathcal{F}_{p'}$ be the stalk functor defined by Equation (04EH) using $u$. There is a canonical comparison map $c : \mathcal{F}_{p'} \to \mathcal{F}_p = p^{-1}\mathcal{F}$. Namely, given a triple $(U, x, \sigma)$ representing an element $\xi$ of $\mathcal{F}_{p'}$ we think of $\sigma$ as a map $\sigma : h^\#_U \to \mathcal{F}$ and we can set $c(\xi) = p^{-1}(\sigma)(x)$ since $x \in u(U) = p^{-1}(h^\#_U)$. By Lemma 00Y6 we see that $(h_U)_{p'} = u(U)$. Since conditions (1) and (2) of Definition 00Y5 hold for $p'$ we also have $(h^\#_U)_{p'} = (h_U)_{p'}$ by Lemma 00Y8. Hence we have

$$(h^\#_U)_{p'} = (h_U)_{p'} = u(U) = p^{-1}(h^\#_U)$$

We claim this bijection equals the comparison map $c : (h^\#_U)_{p'} \to p^{-1}(h^\#_U)$ (verification omitted). Any sheaf on $\mathcal{C}$ is a coequalizer of maps of coproducts of sheaves of the form $h^\#_U$, see Lemma 00WS. The stalk functor $\mathcal{F} \mapsto \mathcal{F}_{p'}$ and the functor $p^{-1}$ commute with arbitrary colimits (as they are both left adjoints). We conclude $c$ is an isomorphism for every sheaf $\mathcal{F}$. Thus the stalk functor $\mathcal{F} \mapsto \mathcal{F}_{p'}$ is isomorphic to $p^{-1}$ and we in particular see that it is exact. This proves condition (3) of Definition 00Y5 holds and $p'$ is a point. The final assertion has already been shown above, since we saw that $p^{-1} = (p')^{-1}$. □